

DIFFERENTIAL EQUATIONS LECTURE NOTES

WEEK IV

REVIEW OF MATRICES

Definition A matrix is an array of numbers

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1j} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2j} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{i1} & a_{i2} & \dots & a_{ij} & \dots & a_{in} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mj} & \dots & a_{mn} \end{bmatrix}$$

$\uparrow$ # of columns.
 $\uparrow$ a_{ij} -th entry of A.
 $\downarrow$ # of rows
 $\downarrow$ number of columns.
 $\downarrow$ Size of matrix (m by n)
 $\downarrow$ number of rows

Ex 7 $A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & -1 & 2 \end{bmatrix}_{2 \times 3}$ $a_{23} = 2$
 $a_{12} = 2$

Definition $A = (a_{ij})_{m \times n}$

- If $a_{ij} \in \mathbb{R} \Rightarrow A$ is called "real matrix" ex $A = \begin{bmatrix} 1 & -2 \\ 10 & 3 \end{bmatrix}$ real matrix
- If $a_{ij} \in \mathbb{C}$ (complex numbers) $\Rightarrow A$ is called "Complex matrix"
 Ex $A = \begin{bmatrix} i & 1+2i \\ -1 & i-2 \end{bmatrix}$ complex matrix

Complex Numbers

$$z = a + bi$$

$\downarrow$ Complex number
 $\downarrow$ a $\downarrow$ b
 $\searrow$ $\text{Re}(z)$: real part of z
 $\searrow$ $\text{Im}(z)$: imaginary part of z

$$\bar{z} = a - bi$$

conjugate of z

$$z \cdot \bar{z} = (a + bi)(a - bi) = a^2 - abi + abi - b^2 \underbrace{\begin{pmatrix} i^2 \\ -1 \end{pmatrix}}_{-1} = a^2 + b^2 = |z|^2$$

$\downarrow$
 Mod of $z = |z|$

Properties of Matrices

$$A = [a_{ij}]_{m \times n}$$

① Transpose of A $A^T = [a_{ji}]_{n \times m}$

Ex $A = \begin{bmatrix} 1 & 2 \\ -1 & 0 \\ 3 & 5 \end{bmatrix}_{3 \times 2} \Rightarrow A^T = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 5 \end{bmatrix}_{2 \times 3}$

② Complex Conjugate of A $\bar{A} = [\bar{a}_{ij}]$

Ex $A = \begin{bmatrix} i & 0 \\ -1 & 2-i \end{bmatrix} \quad \bar{A} = \begin{bmatrix} -i & 0 \\ -1 & 2+i \end{bmatrix}$

③ Adjoint of A $A^* = \bar{A}^T$

$$A^* = \bar{A}^T = \begin{bmatrix} -i & 0 \\ -1 & 2+i \end{bmatrix}^T = \begin{bmatrix} -i & -1 \\ 0 & 2+i \end{bmatrix}$$

④ Equality $A_{m \times n} = B_{m \times n} \quad a_{ij} = b_{ij} \quad \forall i, j$ (their sizes must be equal)

⑤ Zero matrix $A = (a_{ij}) \quad a_{ij} = 0 \quad \forall i, j \quad A = \begin{bmatrix} 0 & \dots & 0 \\ 0 & \dots & 0 \end{bmatrix}$

⑥ Identity matrix $I_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ 0 & 0 & \dots & 1 \\ 0 & 0 & \dots & 0 \end{bmatrix}_{n \times n}$ $a_{ii} = 1$
 $a_{ij} = 0 \quad i \neq j$
 $I_{n \times n} = I_n$
main diagonal

⑦ Square Matrix A is a square matrix if # rows = # columns.

⑧ Addition $A = (a_{ij})_{m \times n}, B = (b_{ij})_{m \times n}$ (A, B must have same size)

$$A \mp B = (a_{ij} \mp b_{ij})_{m \times n}$$

$$A+B = B+A$$

$$A = \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix}_{2 \times 2} \quad B = \begin{bmatrix} 0 & -2 \\ 2i & i \end{bmatrix}_{2 \times 2}$$

$$A+B = \begin{bmatrix} -1 & -2 \\ 2+2i & 3+i \end{bmatrix}_{2 \times 2}$$

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⑨ Multiplication with a scalar C

$$CA = C(a_1 \vec{e}_1) = C(a_1 \vec{e}_1) \quad A = \begin{bmatrix} 1 & 1 \\ 0 & -2 \end{bmatrix} \Rightarrow 2A = \begin{bmatrix} 2 & 2 \\ 0 & -4 \end{bmatrix}$$

⑩ Multiplication of two matrices

$$[A]_{m \times n} [B]_{n \times p} = [AB]_{m \times p} = [C]_{m \times p}$$

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

- $A \cdot B \neq B \cdot A$ (in general)
- $A(BC) = (AB)C$
- $A(B+C) = AB + AC$

$$\text{Ex } A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \end{bmatrix}_{2 \times 3}, B = \begin{bmatrix} 2 & 1 \\ -1 & 0 \\ 0 & 3 \end{bmatrix}_{3 \times 2}$$

$$A \cdot B = \begin{bmatrix} 1 \cdot 2 + 2 \cdot (-1) + (-1) \cdot 0 & 1 \cdot 1 + 0 \cdot 2 + (-1) \cdot 3 \\ 0 \cdot 2 + (-1) \cdot 3 + 1 \cdot 0 & 0 \cdot 1 + 0 \cdot 3 + 1 \cdot 3 \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ -3 & 3 \end{bmatrix}$$

$$[A]_{m \times n} \cdot [B]_{n \times p} = [AB]_{m \times p}$$

⑪ Multiplication of vectors

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1} : \text{column vector}$$

$$x = \underbrace{[x_1 \ x_2 \ \dots \ x_n]}_{\text{row vector}}_{1 \times n}$$

$$\text{Ex } x = [1, 2, 3] \\ y = [0, -1, 0]$$

$$x \cdot y = 1 \cdot 0 + 2 \cdot (-1) + 3 \cdot 0 = -2$$

• Inner (Dot) Product (Scalar Product)

$$x = [x_1 \ x_2 \ \dots \ x_n] \\ y = [y_1 \ y_2 \ \dots \ y_n]$$

$$x \cdot y = x \cdot y^T = x_1 y_1 + \dots + x_n y_n \\ = \sum_{i=1}^n x_i y_i$$

when x, y real vectors
 $x \cdot y = y \cdot x$

$$\bullet \text{ If } x, y \in \mathbb{C} \Rightarrow x \cdot y = \sum_{i=1}^n x_i \bar{y}_i$$

$$\bullet \text{ If } x \cdot y = 0 \Rightarrow x \perp y \text{ (they are perpendicular (orthogonal) to each other)}$$

Norm of x

$$\|x\| = \sqrt{x \cdot x}$$

$$\text{Ex } x = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, y = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \quad x \cdot y = (x, y) = 1 \cdot (-1) + 1 \cdot (1) = 0$$

12) Inverse of A A is square, $n \times n$

A matrix $A_{n \times n}$ has an inverse if there is a matrix $B_{n \times n}$ such that $A \cdot B = B \cdot A = I_n \rightarrow$ identity matrix

- Such a matrix is called **invertible (nonsingular)**. The matrix B is called **inverse of A**, and is denoted by $B = A^{-1}$
- If A has NO inverse, it is called **singular** or **noninvertible**

FINDING A^{-1}

1) Gaussian Elimination (By using elementary operations)

Elementary Row Operations

- Interchanging two rows: $R_i \leftrightarrow R_j$ $A = \begin{bmatrix} 1 & 2 \\ -3 & -1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} -3 & -1 \\ 1 & 2 \end{bmatrix}$
- Multiplying a row with a constant c : cR_i $A \xrightarrow{2R_2} \begin{bmatrix} 1 & 2 \\ -6 & -2 \end{bmatrix}$
- Multiplying a row with a constant $c \neq 0$, and add it to another row: $cR_i + R_j$ $A \xrightarrow{2R_1 + R_2} \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$

To find A^{-1} : write augmented matrix

$$\left[A_{n \times n} \mid I_{n \times n} \right] \xrightarrow{\substack{\text{Apply row operations} \\ \text{to convert } A \text{ into } I}} \left[I \mid A^{-1} \right]$$

$\downarrow$ identity

$$\text{Ex } \xrightarrow{\quad} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ -3 & -1 & 0 & 1 \end{array} \right] \xrightarrow{3R_1 + R_2} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 7 & 3 & 1 \end{array} \right] \xrightarrow{\frac{1}{7}R_2} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & 3/7 & 1/7 \end{array} \right]$$

$$\xrightarrow{-2R_2 + R_1} \left[\begin{array}{cc|cc} 1 & 0 & 1/7 & -2/7 \\ 0 & 1 & 3/7 & 1/7 \end{array} \right]$$

$\underbrace{\quad}_{I} \quad \underbrace{\quad}_{A^{-1}}$

$$\text{So } A^{-1} = \begin{bmatrix} 1/7 & -2/7 \\ 3/7 & 1/7 \end{bmatrix}$$

② Using Determinant, $A_{n \times n}$ square

$$n=1, \quad A = [a_{11}] \Rightarrow \det(A) = |A| = a_{11}$$

$$n=2, \quad A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \Rightarrow \det(A) = |A| = a_{11}a_{22} - a_{12}a_{21}$$

$$n=3, \quad A = \begin{bmatrix} + & - & + \\ a_{11} & a_{12} & a_{13} \\ - & + & - \\ a_{21} & a_{22} & a_{23} \\ + & - & + \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \Rightarrow \det(A) = |A| = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\underline{n \times n} \quad \det(A) = |A| = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(A_{ij})$$

the matrix obtained from A by deleting i -th row and j -th column

$$\text{Ex } \begin{vmatrix} + & - & + \\ 1 & 2 & 0 \\ - & + & - \\ -1 & 0 & 1 \\ + & - & + \\ 3 & 1 & -2 \end{vmatrix} = 1 \begin{vmatrix} 0 & 1 \\ 1 & -2 \end{vmatrix} - 2 \begin{vmatrix} -1 & 1 \\ 3 & -2 \end{vmatrix} + 0 \begin{vmatrix} -1 & 0 \\ 3 & 1 \end{vmatrix} = -1 - 2(2-3) + 0 = 1$$

Theorem Let A be $n \times n$ matrix. Then, A^{-1} exists iff $\det A \neq 0$

Definition $A^{-1} = \frac{1}{\det A} C^T$ $\rightarrow$ adjoint of A .

where $C_{ij} = [(-1)^{i+j} \det A_{ij}]$ with A_{ij} is the matrix obtained by deleting the i -th row and j -th column of the original matrix

$$\text{Ex } A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \quad \det A = (-2-1) - (4-1) - (2+1) = -3-3-3 = -9 \neq 0$$

so A^{-1} exists

$$C_{11} = (-1)^{1+1} \det A_{11} \text{ where } |A_{11}| = \begin{vmatrix} -1 & 1 \\ 1 & 2 \end{vmatrix} = -3 \Rightarrow C_{11} = -3$$

$$C_{12} = (-1)^{1+2} \det A_{12} \quad |A_{12}| = (-1)^{1+2} \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = -3$$

$$C_{13} = (-1)^{1+3} \det A_{13} \quad |A_{13}| = (-1)^{1+3} \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} = 3$$

Thus, $C = \begin{bmatrix} -3 & -3 & 3 \\ -3 & 3 & 0 \\ 0 & -3 & 3 \end{bmatrix}$ with $C^T = \begin{bmatrix} -3 & -3 & 0 \\ -3 & 3 & -3 \\ 3 & 0 & 3 \end{bmatrix}$

⑤

$$\text{Then, } A^{-1} = \frac{1}{\det A} C^T = \frac{1}{-9} \begin{bmatrix} -3 & -3 & 0 \\ -3 & 3 & -3 \\ 3 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1/3 & 1/3 & 0 \\ 1/3 & -1/3 & 1/3 \\ -1/3 & 0 & -1/3 \end{bmatrix}$$

Remark $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad A^{-1} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Effects of Row Operation on $\det A$

- ① $cR_i \rightarrow R_i \quad (c \neq 0) \Rightarrow$ multiplies $\det A$ by c
- ② $R_i \leftrightarrow R_j \quad (i \neq j) \Rightarrow$ multiplies $\det A$ by (-1)
- ③ $cR_i + R_j \rightarrow R_j \Rightarrow$ doesn't change $\det A$.

Note $\det(cA) = c^n \det A$ if A is of size $n \times n$.

Ex $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} \quad \det A = 3 + 2 = 5.$

$$A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} \xrightarrow{2R_1} \begin{bmatrix} 2 & -2 \\ 2 & 3 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 2 & 3 \\ 2 & -2 \end{bmatrix} \xrightarrow{2R_1 + R_2} \begin{bmatrix} 2 & 3 \\ 6 & 4 \end{bmatrix} =$$

$\det A \quad \quad \quad 2 \cdot \det A \quad \quad \quad -2 \det A$

$$\det B = -2 \cdot \det A = -2 \cdot 5 = -10$$

$$\det(B) = 2 \cdot 4 - 6 \cdot 3 = -10$$

(B) MATRIX FUNCTIONS

Matrices of which entries are functions

$$A = \begin{bmatrix} e^{3t} & t \sin t & 1 \\ 2t & e^{2t} & \cos t \end{bmatrix}$$

$$\frac{dA}{dt} = \frac{d[A_{ij}(t)]}{dt} = \begin{bmatrix} 3e^{3t} & \ln t + t \cdot \frac{1}{t} & 0 \\ 2 & 2e^{2t} & -\sin t \end{bmatrix}$$

if $a_{ij}(t)$ is
differentiable
and
integrable

$$\int A = \left[\int [a_{ij}(t)] dt \right] = \begin{bmatrix} \frac{e^{3t}}{3} & \int t \sin t dt & t \\ t^2 & \frac{e^{2t}}{2} & \sin t \end{bmatrix}$$

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14 TRIANGULAR MATRICES

• Diagonal matrices

Ex $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

$$\begin{bmatrix} x & & & 0 \\ & x & & \\ & & x & \\ 0 & & & x \end{bmatrix}$$

$$a_{pp} \neq 0$$

$$a_{ij} = 0$$

$$\text{for } i \neq j$$

Diagonal matrices
n x n matrices

• Upper Triangular Matrices

Upper-Triangular matrices
n x n matrices

$$\begin{bmatrix} x & & & \\ & x & & \\ & & x & \\ 0 & & & x \end{bmatrix}$$

$$a_{ij} = 0 \quad i > j \quad \text{Ex } \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$$

• Lower Triangular Matrices

Lower Triangular
Matrices n x n matrices

$$\begin{bmatrix} x & & 0 \\ & x & \\ & & x & \\ x & & & x \end{bmatrix}$$

$$a_{ij} = 0 \quad i < j$$

$$\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$$

SYSTEMS OF LINEAR ALGEBRAIC OPERATIONS

"System of linear equations
in n-unknowns"

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases}$$

m-equations
n-unknowns
 $x_1, x_2, \dots, x_n$ are unknowns

This can be written as

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

A: coefficient matrix

$\vec{x}$: unknown vector

$\vec{b}$: right hand side vector

$$A\vec{x} = \vec{b}$$

$\longrightarrow$
A is invertible

$$A^{-1}A\vec{x} = A^{-1}\vec{b}$$

$$\vec{I} \cdot \vec{x} = A^{-1}\vec{b}$$

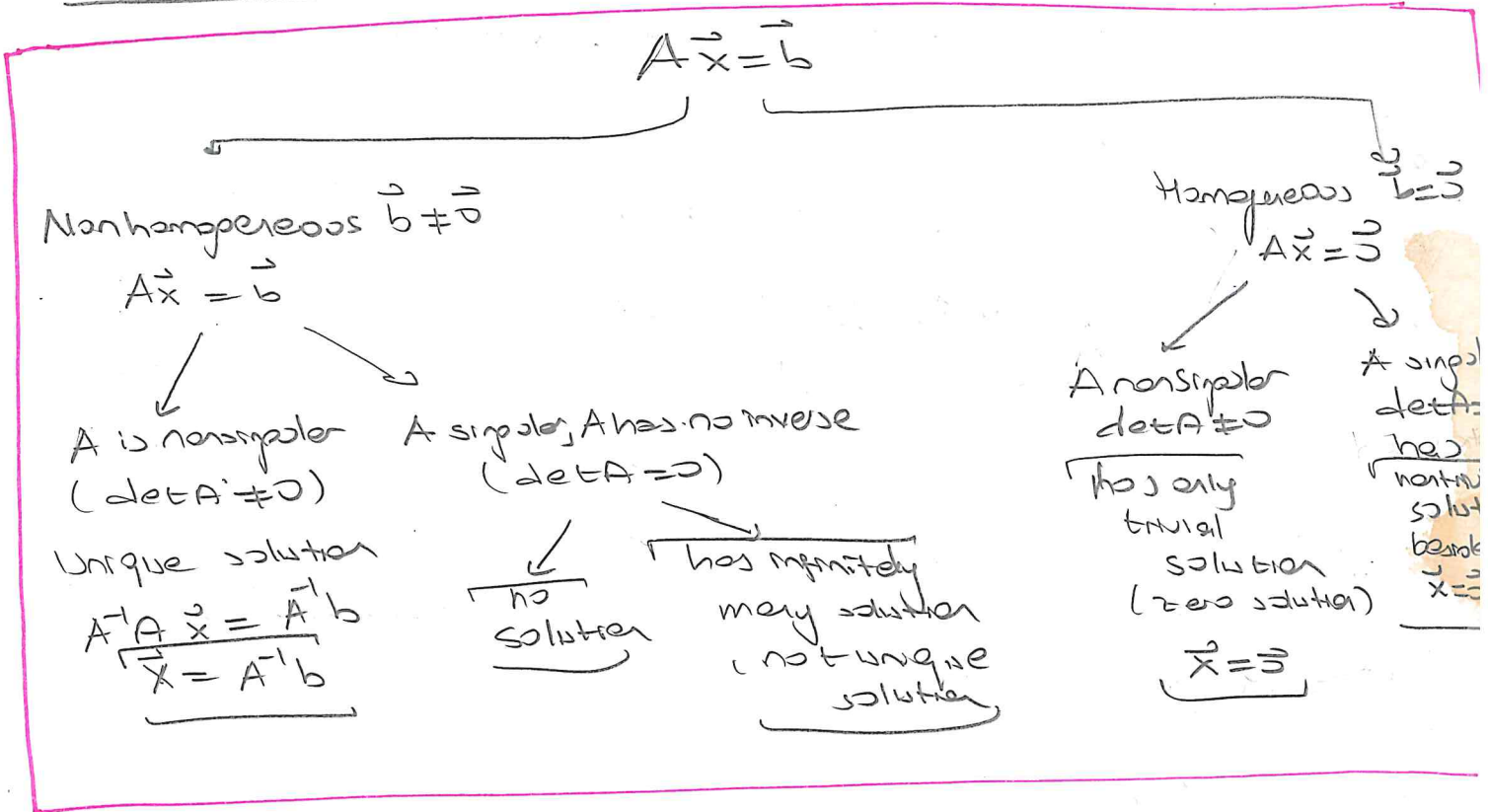
$$\vec{x} = A^{-1}\vec{b}$$

if $\vec{b} = 0$
 $\vec{x} = 0$

Definition If $b=0$, then $A\vec{x}=\vec{0}$ and it is called

homogeneous otherwise, $A\vec{x}=\vec{b}$ is called nonhomogeneous.

For square systems, $A_{n \times n}$, we have following:



→ The solution of the $A\vec{x}=\vec{b}$ can be found by using

Gaussian elimination using the augmented matrix

$$[A|\vec{b}] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right] \xrightarrow[\text{operations}]{\text{apply row}} \left(\begin{array}{c} \text{upper} \\ \text{triangular} \\ \text{matrix} \end{array} \middle| \vec{b} \right)$$

use backward substitution to obtain solution

Ex: $x_1 - x_3 + x_4 = 0$

$3x_1 + x_2 + x_3 = 1$

$-x_1 + x_2 + 2x_3 - x_4 = 2$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 3 & 1 & 1 & 0 & 1 \\ -1 & 1 & 2 & -1 & 2 \end{array} \right] \xrightarrow[\substack{-3e_1+e_2 \\ R_1+R_3}]{\substack{-3e_1+e_2 \\ R_1+R_3}} \left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 4 & -3 & 1 \\ 0 & 1 & 1 & 0 & 2 \end{array} \right] \xrightarrow{-R_2+R_3} \left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 4 & -3 & 1 \\ 0 & 0 & -3 & 3 & 1 \end{array} \right]$$

$$R_3 \rightarrow -\frac{1}{3}R_3 \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 4 & -3 & 1 \\ 0 & 0 & 1 & -1 & -\frac{1}{3} \end{array} \right]$$

Leading 1's (leading variable: the first non-zero entry of each row is called leading variable)

The columns that do not contain a leading 1 correspond to free variables.

Here, x_4 : free variable

x_1, x_2, x_3 : are leading unknowns. This means that x_1, x_2, x_3 can be written in terms of x_4 .

Then, the last equation will be.

$$x_1 - x_3 + x_4 = 0 \Rightarrow x_1 = x_3 - x_4 = x_4 - \frac{1}{3} - x_4 = -\frac{1}{3}$$

$$x_2 + 4x_3 - 3x_4 = 1 \Rightarrow x_2 = 1 + 3x_4 - 4x_3 = 1 + 3x_4 - 4(x_4 - \frac{1}{3}) = \frac{7}{3} - x_4$$

$$x_3 - x_4 = -\frac{1}{3} \Rightarrow x_3 = x_4 - \frac{1}{3}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} \\ \frac{7}{3} - x_4 \\ x_4 - \frac{1}{3} \\ x_4 \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} \\ \frac{7}{3} \\ -\frac{1}{3} \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \end{bmatrix}$$

So we have infinitely many solutions

Definition # leading entries (1's) is called the rank of a matrix A. Thus, in previous example $\text{rank}(A)=3$

Ex

$$\begin{aligned} x_1 - x_2 + 3x_3 + x_4 + x_5 &= -1 \\ x_1 - 2x_3 + x_4 + 2x_5 &= 3 \\ 2x_1 - 2x_2 + 6x_3 + 4x_5 &= 2 \end{aligned}$$

$$\left[\begin{array}{ccccc|c} 1 & -1 & 3 & 1 & 1 & -1 \\ 1 & 0 & -2 & 1 & 2 & 3 \\ 2 & -2 & 6 & 0 & 4 & 2 \end{array} \right] \xrightarrow{\substack{-R_1+R_2 \\ -2R_1+R_3}} \left[\begin{array}{ccccc|c} 1 & -1 & 3 & 1 & 1 & -1 \\ 0 & 1 & -5 & 0 & 1 & 4 \\ 0 & 0 & 0 & -2 & 2 & 4 \end{array} \right]$$

$$\xrightarrow{-\frac{1}{2}R_3} \left[\begin{array}{ccccc|c} 1 & -1 & 3 & 1 & 1 & -1 \\ 0 & 1 & -5 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 & -1 & -2 \end{array} \right]$$

leading variable

x_1, x_2, x_4 : leading 1's $\Rightarrow \text{rank } A = 3$, so x_1, x_2, x_4 can be written in terms of x_3, x_5 .

x_3, x_5 : free variables

$$\begin{aligned} x_1 - x_2 + 3x_3 + x_4 + x_5 &= -1 \Rightarrow x_1 = -1 + x_2 - 3x_3 - x_4 - x_5 = 5 + 2x_3 - 3x_5 \\ x_2 - 5x_3 + x_5 &= 4 \Rightarrow x_2 = 4 + 5x_3 - x_5 \\ x_4 - x_5 &= -2 \Rightarrow x_4 = -2 + x_5 \end{aligned}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 5 + 2x_3 - 3x_5 \\ 4 + 5x_3 - x_5 \\ x_3 \\ -2 + x_5 \\ x_5 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 0 \\ -2 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 5 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

"infinitely many solutions"

Remark In Gaussian elimination, we aim to convert to given matrix A into a new matrix, which is called row reduced echelon form of the A .

Definition A matrix satisfying the following properties is called row reduced echelon form.

- ① The first non-zero entry of each row is 1, which are called leading 1's.
- ② All entries which are below a leading 1 are 0.
- ③ If $i < j$ then the leading 1 on row i must be to the left of the leading 1 on row j .
- ④ If there are k rows of 0's in the matrix, they must be at the bottom.

Ex 7

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & -5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

leading 1's
"row-reduced echelon"

$$\begin{bmatrix} 1 & 5 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

leading 1's
"row reduced echelon"

$$\begin{bmatrix} 3 & 2 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

not row-reduced echelon

$$\begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2}$$

row reduced echelon

Thm Any matrix can be reduced until a matrix in row echelon form is obtained.

Ex 7 ①

$$\begin{bmatrix} 3 & 2 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{1/3 R_1} \begin{bmatrix} 1 & 2/3 & -1/3 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

leading 1's, not free variable
row reduced echelon

②

$$\begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}$$

"row-reduced echelon"
leading 1's → one free variable
(1 1)

LINEAR INDEPENDENCE

Definition If $\vec{x}^{(1)}, \vec{x}^{(2)}, \dots, \vec{x}^{(n)}$ be vectors, then

$$c_1 \vec{x}^{(1)} + c_2 \vec{x}^{(2)} + \dots + c_n \vec{x}^{(n)}$$

is called the linear combination of $\vec{x}^{(1)}, \vec{x}^{(2)}, \dots, \vec{x}^{(n)}$ for some nonzero constant $c_i, i=1, 2, \dots, n$

Ex $\vec{x}^{(1)} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{x}^{(2)} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \Rightarrow 2\vec{x}^{(1)} - 3\vec{x}^{(2)}$ is a linear combination of $\vec{x}^{(1)}$ and $\vec{x}^{(2)}$.

Definition A set of n vector $\vec{x}^{(1)}, \vec{x}^{(2)}, \dots, \vec{x}^{(n)}$ is called linearly independent if

$$c_1 \vec{x}^{(1)} + c_2 \vec{x}^{(2)} + \dots + c_n \vec{x}^{(n)} = \vec{0}$$

Ex $\vec{x}^{(1)} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $\vec{x}^{(2)} = \begin{bmatrix} -3 \\ -6 \end{bmatrix}$

$$\vec{x}^{(2)} = -3\vec{x}^{(1)} \quad \vec{x}^{(3)} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

implies that $c_1 = c_2 = \dots = c_n = 0$ otherwise, it is called

linearly dependent.

In other words, a set of vectors is linearly independent iff none of the vectors can be expressed as a linear combination of remaining vectors.

Let $\vec{x}^{(p)} = \begin{bmatrix} x_1^{(p)} \\ x_2^{(p)} \\ \vdots \\ x_n^{(p)} \end{bmatrix}$, then $c_1 \vec{x}^{(1)} + c_2 \vec{x}^{(2)} + \dots + c_n \vec{x}^{(n)} = \vec{0}$ will give us the following linear homogeneous system:

$$c_1 \begin{bmatrix} x_1^{(1)} \\ x_2^{(1)} \\ \vdots \\ x_n^{(1)} \end{bmatrix} + c_2 \begin{bmatrix} x_1^{(2)} \\ x_2^{(2)} \\ \vdots \\ x_n^{(2)} \end{bmatrix} + \dots + c_n \begin{bmatrix} x_1^{(n)} \\ x_2^{(n)} \\ \vdots \\ x_n^{(n)} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1^{(1)} & x_1^{(2)} & \dots & x_1^{(n)} \\ x_2^{(1)} & x_2^{(2)} & \dots & x_2^{(n)} \\ \vdots & \vdots & \ddots & \vdots \\ x_n^{(1)} & x_n^{(2)} & \dots & x_n^{(n)} \end{bmatrix} \underbrace{\begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}}_{\text{Unknown}} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \Rightarrow \vec{X} \vec{c} = \vec{0} \quad \text{"homogeneous linear system with unknown } \vec{c} \text{"}$$

Conclusion 1 If $\det \vec{X} \neq 0 \Rightarrow \vec{X}$ is invertible and the system has only the trivial (zero) solution. That is,

$$\vec{c} = \vec{0} \quad (\text{i.e. } c_p = 0 \quad \forall p=1, 2, \dots, n)$$

Thus, the vectors are linearly independent.

2 If $\det \vec{X} = 0 \Rightarrow$ the system has non-zero solutions.

That is $c_p \neq 0$ for some p . Thus, the vectors $\vec{x}^{(1)}, \dots, \vec{x}^{(n)}$ are linearly dependent.

Remark Thus, $\vec{x}^{(1)}, \vec{x}^{(2)}, \dots, \vec{x}^{(n)}$ are linearly independent iff $\det \vec{X} \neq 0$

Ex Determine the vectors $\vec{x} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$, $\vec{y} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$, $\vec{z} = \begin{pmatrix} -4 \\ 1 \\ -11 \end{pmatrix}$ are linearly independent or not.

Construct the system $[\vec{x} \ \vec{y} \ \vec{z}] \cdot \vec{c} = \vec{0}$ such that

$$\underbrace{\begin{bmatrix} 1 & 2 & -4 \\ 2 & 1 & 1 \\ -1 & 3 & -11 \end{bmatrix}}_X \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{"solve this hom. system!"}$$

det $X = (-11-3) - 2(-22+1) - 4(6+1) = -14 + 42 - 28 = 0 \Rightarrow$
the system has non-zero solutions, that is, some $c_i \neq 0$.

Thus the vectors are linearly dependent, that is, vectors can be written as linear combinations of others.

Find these combinations by solving the system.

$$\left[\begin{array}{ccc|c} 1 & 2 & -4 & 0 \\ 2 & 1 & 1 & 0 \\ -1 & 3 & -11 & 0 \end{array} \right] \xrightarrow[\substack{R_1+R_3 \\ \text{side vector} \\ \text{not need} \\ \text{to write}}]{\substack{-2R_1+R_2 \\ R_1+R_3}} \left[\begin{array}{ccc|c} 1 & 2 & -4 & 0 \\ 0 & -3 & 9 & 0 \\ 0 & 5 & -15 & 0 \end{array} \right] \xrightarrow[\substack{1/5 R_3}]{-1/3 R_2} \left[\begin{array}{ccc|c} 1 & 2 & -4 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 1 & -3 & 0 \end{array} \right]$$

$$\xrightarrow{-2R_2+R_3} \left[\begin{array}{ccc|c} 1 & 2 & -4 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Leading is c_3 is free variable.

(c_1 & c_2 can be written in terms of c_3)

$$c_1 + 2c_2 - 4c_3 = 0 \Rightarrow c_1 = -2c_2 + 4c_3$$

$$c_2 - 3c_3 = 0 \Rightarrow c_2 = 3c_3$$

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = c_3 \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}$$

$$\text{Let } c_3 = 1 \Rightarrow -2\vec{x} + 3\vec{y} + \vec{z} = \vec{0}.$$

Ex. $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ check linear independence $c_1 \vec{x}^{(1)} + c_2 \vec{x}^{(2)} + c_3 \vec{x}^{(3)} = \vec{0}$

$$c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \underbrace{\begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}}_{\vec{X}} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

First way

① $\det \vec{X} = 1 + 1 - 1 = 1 \neq 0$ so the system has only trivial (zero) solution [in other words $\det X \neq 0 \Rightarrow X$

is invertible $X^{-1} X \cdot \vec{c} = X^{-1} \cdot \vec{0} \Rightarrow \vec{c} = \vec{0}$]

Thus, $c_1 = c_2 = c_3 = 0 \Rightarrow$ vectors are linearly independent

Second way

② Solve the system by Gaussian elimination (long way)

$$\begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \xrightarrow[-R_1+R_3]{-R_1+R_2} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{-2R_3+R_2} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 0 & -3 \\ 0 & 1 & 2 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & -3 \end{bmatrix} \xrightarrow{-1/3 R_3} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{cases} c_1 - c_2 - c_3 = 0 \\ c_2 + 2c_3 = 0 \\ c_3 = 0 \end{cases}$$

So, $c_2 = 0, c_3 = 0$
 $c_1 = c_2 = c_3 = 0 \Rightarrow$ vectors are linearly independent.

Definition A set of functions $\{f_1(t), f_2(t), \dots, f_n(t)\}$ is linearly independent if the equation

$$c_1 f_1(t) + c_2 f_2(t) + \dots + c_n f_n(t) = 0 \text{ for all } t.$$

implies $c_1 = c_2 = \dots = c_n = 0$.

Ex 7 Show that $e^{r_1 t}$ and $e^{r_2 t}$, $r_1, r_2 \in \mathbb{R}$ and $r_1 \neq r_2$ are linearly independent

$$c_1 e^{r_1 t} + c_2 e^{r_2 t} = 0 \quad \text{for all } t.$$

Choose some specific values of t to obtain a linear system

$$t=0 \Rightarrow c_1 + c_2 = 0$$

$$t=1 \Rightarrow c_1 e^{r_1} + c_2 e^{r_2} = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 1 \\ e^{r_1} & e^{r_2} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\det X = e^{r_2} - e^{r_1} = e^{r_2} (1 - e^{r_1 - r_2}) \neq 0$$

$\neq 1$ since $r_1 \neq r_2$
 $\neq 0$

So, $c_1 = c_2 = 0 \Rightarrow e^{r_1 t}$ and $e^{r_2 t}$ are linearly independent

Remark There is another way to check linear independence of functions $\{f_1(t), f_2(t), \dots, f_n(t)\}$ if $f_i(t)$ are differentiable enough times.

$$c_1 f_1 + c_2 f_2 + \dots + c_n f_n = 0 \quad \text{holds for all } t.$$

Differentiate the equation repeatedly in order to construct a linear system:

$$c_1 f_1 + c_2 f_2 + \dots + c_n f_n = 0$$

$$c_1 f_1' + c_2 f_2' + \dots + c_n f_n' = 0$$

$$c_1 f_1'' + c_2 f_2'' + \dots + c_n f_n'' = 0$$

$$\vdots$$

$$c_1 f_1^{(n-1)} + c_2 f_2^{(n-1)} + \dots + c_n f_n^{(n-1)} = 0$$

$$\begin{bmatrix} f_1 & f_2 & \dots & f_n \\ f_1' & f_2' & \dots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \dots & f_n^{(n-1)} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

"homogeneous linear system"

$\det F \neq 0 \Rightarrow F$ is invertible and

Definition $\det T = W(f_1, f_2, \dots, f_n) = \begin{vmatrix} f_1 & f_2 & \dots & f_n \\ f_1' & f_2' & \dots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \dots & f_n^{(n-1)} \end{vmatrix}$

is called Wronskian of $f_1, f_2, \dots, f_n$.

Thm If the Wronskian of a set of functions is NOT zero even only for one value of t , then the set is linearly independent. In other words,

$W(f_1, f_2, \dots, f_n) \neq 0$ for some $t \Rightarrow f_1, f_2, \dots, f_n$ are linearly independent

Ex $f_1 = e^{r_1 t}, f_2 = e^{r_2 t} \quad r_1 \neq r_2 \in \mathbb{R}$.

$W(f_1, f_2) = \begin{vmatrix} e^{r_1 t} & e^{r_2 t} \\ r_1 e^{r_1 t} & r_2 e^{r_2 t} \end{vmatrix} = \underbrace{e^{(r_1+r_2)t}}_{\neq 0} (r_2 - r_1) \neq 0 \quad \forall t \Rightarrow f_1, f_2$
 $r_1 \neq r_2$ are linearly independent

Ex $f_1(t) = \cos t, f_2(t) = \sin t$

$W(f_1, f_2) = \begin{vmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{vmatrix} = \cos^2 t + \sin^2 t = 1 \neq 0 \quad \forall t$ so

$\{\cos t, \sin t\}$ are linearly independent.

Ex $f_1 = 1, f_2 = x, f_3 = x^2$

$W(1, x, x^2) = \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & 2x \\ 0 & 0 & 2 \end{vmatrix} = 2 \neq 0 \Rightarrow$ they are lin. independent.

EIGENVALUES & EIGENVECTORS

The linear system

$$[A]_{n \times n} \vec{v} = \lambda \vec{v}$$

where both $\vec{v} \neq \vec{0}$ and scalar $\lambda \in \mathbb{C}$ are unknown,
is called an eigenvalue problem. λ is called **eigenvalue**
of A and $\vec{v} \neq \vec{0}$ is the **corresponding eigenvector**

Rewrite the system as

$$A\vec{v} = \lambda\vec{v}$$

$$A\vec{v} - \lambda\vec{v} = \vec{0}$$

$$(A - \lambda I)\vec{v} = \vec{0} \rightarrow \text{"homogeneous linear system!"}$$

$$\downarrow \text{ for } \vec{v} = \vec{0}$$

Has non-zero solutions if
coefficient matrix is singular.

($\det(A - \lambda I) = 0$), and we
look for non-zero solutions i.e. $\lambda =$

" $\det(A - \lambda I) = 0$ gives an n th degree polynomial
characteristic polynomial"

$\lambda + a_{n-1}\lambda^{n-1} + \dots + a_1\lambda + a_0 = 0$ "roots of this
polynomial are the
eigenvalues of A ."

- Once λ is obtained, we can find the corresponding
eigenvector solving the system

$$(A - \lambda I)\vec{v} = \vec{0} \text{ by inserting the obtained } \lambda \text{ value.}$$

**eigenvector, corresponding
eigenvalue λ .**

Remember Clearly, if $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\det(A - \lambda I) = \begin{vmatrix} a - \lambda & b \\ c & d - \lambda \end{vmatrix} = (a - \lambda)(d - \lambda) - bc = \lambda^2 - \underbrace{(a + d)}_{\text{trace}(A)}\lambda + \underbrace{ad - bc}_{\det A}$$

A second degree polynomial.

Ex 7 Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 2 \\ 0 & -3 \end{bmatrix}$

Eigenvalues

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} -1 - \lambda & 2 \\ 0 & -3 - \lambda \end{vmatrix} = (-1 - \lambda)(-3 - \lambda) = 0 = \lambda^2 + 4\lambda + 3 = 0$$

$$(\lambda + 3)(\lambda + 1) = 0$$

$$\lambda_1 = -3 \quad \lambda_2 = -1$$

$\lambda_1 = -3$ $(A - \lambda I)\vec{v} = \vec{0} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

$$(A + 3I)\vec{v} = \vec{0}$$

$$\begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \xrightarrow{1/2 R_1} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} v_{11} \text{ leading } 1 \\ v_{22} \text{ free} \end{array} \quad v_1 + v_2 = 0 \quad v_1 = -v_2$$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -v_2 \\ v_2 \end{bmatrix} = v_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

we can (without loss of generality) let's choose $\vec{v} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ is the eigenvector corresponding to $\lambda = -3$.

Solution space for $(A + 3I)\vec{v} = \vec{0} \Rightarrow$ spanned $\left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$.

$\lambda_2 = -1$ $(A - \lambda I)\vec{v} = \vec{0} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$
 $\Downarrow \lambda = -1$
 $(A + I)\vec{v} = \vec{0}$

$$\begin{bmatrix} 0 & 2 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 2 \\ 0 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} v_{21} \text{ leading } 1 \\ v_{11} \text{ free} \end{array}$$

$$v_2 = 0 \quad v_1 \text{ free}$$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ 0 \end{bmatrix} = v_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

we can let $v_1 = 1 \Rightarrow \vec{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ eigenvector corresponding $\lambda = -1$

Solution space for $(A + I)\vec{v} = \vec{0} \Rightarrow$ spanned $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$

(*) Note, these eigenvectors are not unique.

Definition A matrix A is called diagonalizable if there exists an invertible P such that

$$\underset{\parallel}{J} = P^{-1}AP \text{ is a diagonal matrix}$$

$$\text{diag}(\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n)$$

where $\lambda_i, i=1, 2, \dots, n$ are eigenvalues of $A_{n \times n}$ and

$P = [\vec{v}_1 \ \vec{v}_2 \ \dots \ \vec{v}_n]$ where $\vec{v}_i, i=1, \dots, n$ are the

eigenvectors corresponding to the eigenvalues $\lambda_i, i=1, 2, \dots, n$

Ex (Previous Ex) A is diagonalizable since $JP = [\vec{v} \ \vec{v}] = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$

$$J = \begin{bmatrix} -3 & 0 \\ 0 & -1 \end{bmatrix} = P^{-1}AP \quad A \rightarrow J. \quad \text{or } P = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

↙ check (exercise) ↘

Ex $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ find eigenvalues of A and corresponding eigenvectors.

$J = \begin{bmatrix} -1 & 0 \\ 0 & -3 \end{bmatrix}$

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} = -(\lambda+1)^2(\lambda-2) = 0$$

↙ exercise ↘

$\lambda_{1,2} = -1 \Rightarrow$ repeated eigenvalue / algebraic multiplicity of $\lambda = -1$ is 2.

$\lambda_3 = 2 \Rightarrow \text{alg}(\lambda=2) = 1.$

$\lambda_3 = 2$ $(A - 2I)\vec{v} = 0 \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$

$$\begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \xrightarrow[\substack{1/2 R_1 + R_2 \\ 1/2 R_1 + R_3}]{1/2 R_1 + R_2} \begin{bmatrix} -2 & 1 & 1 \\ 0 & -3/2 & 3/2 \\ 0 & 3/2 & -3/2 \end{bmatrix} \xrightarrow[\substack{R_2 + R_3 \\ -2/3 R_2}]{R_2 + R_3} \begin{bmatrix} -2 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

v_1, v_2 are not free
 v_3 is free.

$$\begin{aligned} -2v_1 + v_2 + v_3 &= 0 \Rightarrow v_1 = 1/2(v_2 + v_3) = v_3 \\ v_2 - v_3 &= 0 \Rightarrow v_2 = v_3 \end{aligned}$$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} v_3 \\ v_2 \\ v_1 \end{bmatrix} = v_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{Sol. space of } (A - 2I)\vec{v} = 0. \Rightarrow \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

$$\dim V = 1 = \# \text{ of}$$

$$\underline{\lambda = -1} \quad (A + I)\vec{v} = 0$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow[-R_1+R_3]{-R_1+R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} u_1 \text{ leading } 1 \\ u_2, u_3: \text{ free.} \end{array}$$

$$\vec{v} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -u_2 - u_3 \\ u_2 \\ u_3 \end{bmatrix} = u_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + u_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Solution space of } (A + I)\vec{v} = \vec{0}$$

$$= \text{spanned} \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\} \Rightarrow \dim V = 2 = \text{alg}(\lambda = -1)$$

free variables

geometric multiplicity of $\lambda = -1$

$$m(\lambda = -1) = 2$$

$$\text{geometric multiplicity} = \# \dim \text{ of eigenspace}$$

Since $m(\lambda = -1) = \text{alg}(\lambda = -1)$, then A is diagonalizable. That is $\exists$ an invertible matrix P .

$$P = [\vec{v} \quad \vec{u}^{(1)} \quad \vec{u}^{(2)}] = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad \det P \neq 0 \text{ and}$$

$$J = P^{-1}AP = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Ex 7 $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ -3 & 2 & 4 \end{bmatrix}$ Find eigenvalues and eigenvectors of A.

$\det(A - \lambda I) = 0$ "characteristic eqn (polynomial)"

$$\begin{vmatrix} 1-\lambda & 1 & 1 \\ 2 & 1-\lambda & -1 \\ -3 & 2 & 4-\lambda \end{vmatrix} = (2-\lambda)^3 = 0 \Rightarrow \lambda = 2 \text{ of } (\lambda=2) = 3$$

$\lambda = 2 \quad (A - 2I)\vec{v} = 0$
 $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$

$\begin{bmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ -3 & 2 & 2 \end{bmatrix} \xrightarrow{\text{row operations}} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

v_1, v_2 : leading 1's
 v_3 = free

1 free = 1 lin. indep. eigenvector

So A is not diagonalizable

$v_1 = v_2 + v_3 \Rightarrow v_1 = 0$
 $v_2 = -v_3$

$\vec{v} = \begin{bmatrix} 0 \\ -v_3 \\ v_3 \end{bmatrix} = v_3 \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \quad V = \text{span} \left\{ \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\}$

Two missing eigenvectors (which cannot be found by usual way of solving $(A - \lambda I)\vec{v} = 0$)

These are called generalized eigenvectors. The way to obtain

them will be given later, then we explain Jordan forms.